

CGN 5930 Risk and Reliability Mid-Term Exam

Problem (1) 20 points

A factory production line is manufacturing bolts using three machines, A, B and C. Of the total output, machine A is responsible for 25%, machine B for 35% and machine C for the rest. It is known from previous experience with the machines that 5% of the output from machine A is defective, 4% from machine B and 2% from machine C. A bolt is chosen at random from the production line and found to be defective. What is the probability that it came from

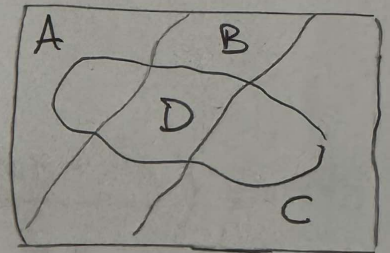
(a) machine A (b) machine B (c) machine C?

$$P(A) = 0.25 \quad P(B) = 0.35 \quad P(C) = 0.4$$

$D = \{ \text{Bolt being defective} \}$

$$P(D|A) = 0.05 \quad P(D|B) = 0.04 \quad P(D|C) = 0.02$$

$$\begin{aligned} P(D) &= \sum_{i=1}^n P(D|E_i) P(E_i) \\ &= P(D|A) P(A) + P(D|B) P(B) + \\ &\quad P(D|C) P(C) \end{aligned}$$



$$= 0.05 \times 0.25 + 0.04 \times 0.35 + 0.02 \times 0.4 = 0.0345$$

$$P(A|D) = \frac{P(D|A) P(A)}{P(D)} = \frac{0.05 \times 0.25}{0.0345} = 0.362$$

$$P(B|D) = \frac{P(D|B) P(B)}{P(D)} = \frac{0.04 \times 0.35}{0.0345} = 0.41$$

$$P(C|D) = \frac{P(D|C) P(C)}{P(D)} = \frac{0.02 \times 0.4}{0.0345} = 0.23$$

Problem (2) 30 points

The annual earthquake design spectral acceleration (S_d) for specific site in Los Anglos is described by a lognormal distribution with $\mu_s = 1.5g$ (where $g=9.81 \text{ m/s}^2$) and coefficient of variation $V_s = 0.25$.

1. What is the design spectral acceleration for a tall building in Los Anglos to be designed to resist 1000-yr earthquake (S_{1000})?
2. What is the probability of that first S_{1000} event to happen within 15 years?
3. What is the probability of having five exceedances of S_{500} in 20 years?

[1]

$$\mu_s = 1.5 \times 9.81 = 14.715 \text{ m/s}^2$$

$$V_s = 0.25$$

$$\lambda_x = \ln \frac{\mu_s}{\sqrt{1+V_s^2}} = \ln \frac{14.715}{\sqrt{1+0.25^2}} = 2.659$$

$$\xi = \sqrt{\ln(1+V_s^2)} = \sqrt{\ln(1+0.25^2)} = 0.246$$

$$P(S \geq S_{1000}) = 0.001$$

$$P(S \leq S_{1000}) = 0.999 \rightarrow \Phi\left(\frac{\ln S_{1000} - \lambda_s}{\xi_s}\right) = 0.999$$

$$\frac{\ln S_{1000} - \lambda_s}{\xi} = \Phi^{-1}(0.999) = 3.08$$

$$\ln S_{1000} = 3.08 \times 0.246 + 2.659 \rightarrow \boxed{S_{1000} = 30.46 \text{ m/s}^2}$$

$$[2] P(X \leq 15) = 1 - (1-p)^x = 1 - (1-0.001)^{15} = 0.015$$

$$[3] P(X=5) = \binom{20}{5} p^x (1-p)^{x-x} = \frac{20!}{5!(20-5)!} (0.002)^5 (0.998)^{15} \\ = 4.81 \times 10^{-10}$$

Problem (3) 20 points

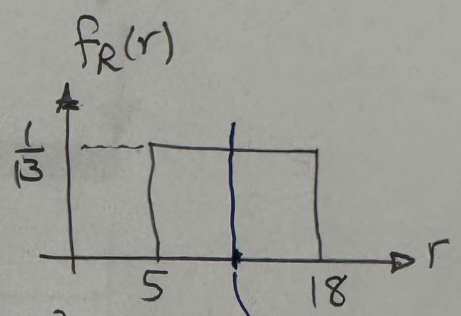
A reinforced concrete beam with moment resistance R is subjected to a bending moment Q . M_R is described by a uniform distribution on the interval from (5,18) while M_Q is described by the below exponential distribution on the interval from (0.4,0.8).

$$f_q(Q) = 2Q^2 + 3Q \cdot \exp(5Q^2)$$

1. What is the Central factor of Safety. (μ_R / μ_Q)

* For the resistance:

$$f_r(R) = \frac{1}{18-5} = \frac{1}{13}$$

$$M_R = E(R) = \int_5^{18} R f_r(R) dR = \frac{R^2}{26} \Big|_5^{18} = \frac{18^2 - 5^2}{26} = 11.5$$


$\swarrow 18-5 = 11.5$

* For the load:

$$M_Q = E(Q) = \int_{0.4}^{0.8} Q f_q(Q) dQ = \int_{0.4}^{0.8} Q (2Q^2 + 3Q \exp(5Q^2)) dQ$$

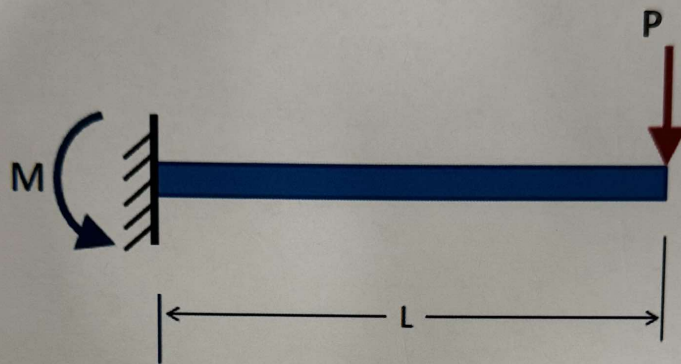
$$= 4.824 \quad \text{"Calculator"}$$

$$\text{Central Factor of Safety} = \frac{M_R}{M_Q} = \frac{11.5}{4.824} = 2.384$$

Problem (4) 30 points

The prismatic cantilever beam in the figure below is loaded with a concentrated load, P , at mid span. The yield moment capacity of the beam is $M_y = 1920$ kips.in where $L = 20$ ft.

1. If the concentrated load P follows a lognormal distribution with mean $\mu = 20$ kips and a coefficient of variation $v = 0.3$, what is the probability of yielding of the beam?
2. What is the 100-year maximum load (P_{100}).



$$\boxed{1} \quad \lambda = \ln \frac{\mu}{\sqrt{1+v^2}} = \ln \frac{20}{\sqrt{1+0.3^2}} = 2.953$$

$$\xi = \sqrt{\ln(1+v^2)} = \sqrt{\ln(1+0.3^2)} = 0.294$$

$$M_y = P_y L \rightarrow P_y = \frac{M_y}{L} = \frac{1920 \times \frac{1}{12}}{20} = 8 \text{ Kips}$$

$$\begin{aligned} P(P \geq P_y) &= P(P > 8) = 1 - P(P < 8) \\ &= 1 - \Phi\left(\frac{\ln 8 - \lambda}{\xi}\right) = 1 - \Phi\left(\frac{\ln 8 - 2.953}{0.294}\right) \end{aligned}$$

$$P(P \geq P_y) = 1 - \Phi(-2.97) = 1 - 0.0015 = 0.9985$$

$$\boxed{2} \quad P(P > P_{100}) = 0.01$$

$$P(P < P_{100}) = 0.99$$

$$\frac{\ln P_{100} - 2.953}{0.294} = \Phi^{-1}(0.99)$$

$$\Phi\left(\frac{\ln P_{100} - \lambda}{\xi}\right) = 0.99$$

$$= 2.33$$

$$\boxed{P_{100} = 38 \text{ Kips}}$$

$$P_{100} = \exp(0.294 \times 2.33 + 2.953) = 38.02 \text{ Kips}$$